

Lie Groups III Examples Bottom Up

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1 $\mathfrak{so}(3)$ Lie Algebra

1.1 Step 1: Define the $\mathfrak{so}(3)$ Lie Algebra

The Lie algebra $\mathfrak{so}(3)$ is a three-dimensional real Lie algebra generated by basis elements J_x, J_y, J_z with the defining commutation relations:

$$[J_i, J_j] = i\epsilon_{ijk}J_k, \quad (1)$$

where $i, j, k = x, y, z$, ϵ_{ijk} is the Levi-Civita symbol (totally antisymmetric with $\epsilon_{xyz} = 1$), and summation over repeated indices is implied. Explicitly, these are:

- $[J_x, J_y] = iJ_z$,
- $[J_y, J_z] = iJ_x$,
- $[J_z, J_x] = iJ_y$.

This convention, common in physics, ensures that in unitary representations, the operators are Hermitian, which is suitable for the ladder operator approach. Alternatively one can absorb the i into the generators and obtain $[J_i, J_j] = \epsilon_{ijk}J_k$.

1.2 Step 2: Introduce Ladder Operators

In the ladder operator method, we define the raising and lowering operators as:

$$J_+ = J_x + iJ_y, \quad J_- = J_x - iJ_y. \quad (2)$$

These operators are complex linear combinations of the generators, facilitating the analysis of representations. To proceed, we compute their commutation relations using the $\mathfrak{so}(3)$ algebra.

$$[J_z, J_+] = [J_z, J_x + iJ_y] = [J_z, J_x] + i[J_z, J_y] = \quad (3)$$

$$= iJ_y + i(-iJ_x) = iJ_y + J_x = J_x + iJ_y = J_+ , \quad (4)$$

$$[J_z, J_-] = [J_z, J_x - iJ_y] = [J_z, J_x] - i[J_z, J_y] = \quad (5)$$

$$= iJ_y - i(-iJ_x) = iJ_y - J_x = -(J_x - iJ_y) = -J_- , \quad (6)$$

$$[J_+, J_-] = [J_x + iJ_y, J_x - iJ_y] = [J_x, J_x] - i[J_x, J_y] + i[J_y, J_x] - i^2[J_y, J_y] = \quad (7)$$

$$= 0 - i(iJ_z) + i(-iJ_z) + (-1) \cdot 0 = -i^2 J_z + i^2 J_z = J_z - (-J_z) = 2J_z . \quad (8)$$

Thus, the ladder operator algebra is:

$$[J_z, J_+] = J_+ , \quad (9)$$

$$[J_z, J_-] = -J_- , \quad (10)$$

$$[J_+, J_-] = 2J_z . \quad (11)$$

1.3 Step 3: Representations of the Lie Algebra

A representation of $\mathfrak{so}(3)$ is a homomorphism from the Lie algebra to the linear operators on a complex vector space V , preserving the commutation relations. Since $\text{SO}(3)$ is compact, its finite-dimensional representations are completely reducible, so we focus on irreducible representations.

Assume V is a finite-dimensional complex vector space where J_z is diagonalizable (possible in complex representations). Let $|m\rangle$ be an eigenvector of J_z with eigenvalue m :

$$J_z|m\rangle = m|m\rangle . \quad (12)$$

• Action of J_+ :

$$J_z(J_+|m\rangle) = (J_+J_z + [J_z, J_+])|m\rangle = J_+J_z|m\rangle + J_+|m\rangle = J_+(m|m\rangle) + J_+|m\rangle = m(J_+|m\rangle) + J_+|m\rangle = (m+1)(J_+|m\rangle) \quad (13)$$

Thus, $J_+|m\rangle$ is either zero or an eigenvector of J_z with eigenvalue $m+1$.

• Action of J_- :

$$J_z(J_-|m\rangle) = (J_-J_z + [J_z, J_-])|m\rangle = J_-J_z|m\rangle - J_-|m\rangle = J_-(m|m\rangle) - J_-|m\rangle = m(J_-|m\rangle) - J_-|m\rangle = (m-1)(J_-|m\rangle) \quad (14)$$

Thus, $J_-|m\rangle$ is either zero or an eigenvector with eigenvalue $m-1$.

Since V is finite-dimensional, the eigenvalues cannot increase or decrease indefinitely. There must exist a highest eigenvalue j such that:

$$J_+|j\rangle = 0, \quad J_z|j\rangle = j|j\rangle, \quad (15)$$

and a lowest eigenvalue $m_{\min}$ such that $J_-|m_{\min}\rangle = 0$.

1.4 Step 4: Construct the Representation

Starting from the highest weight vector $|j\rangle$, apply J_- repeatedly:

- $J_-|j\rangle$ has eigenvalue $j-1$,
- $J_-^2|j\rangle$ has eigenvalue $j-2$,
- and so on, until $J_-^k|j\rangle \neq 0$ but $J_-^{k+1}|j\rangle = 0$.

Define $|m\rangle \propto J_-^{j-m}|j\rangle$ (with appropriate normalization), where $m = j, j-1, \dots$, until $J_-|m_{\min}\rangle = 0$. Suppose $m_{\min} = -j$, so:

$$J_-|-j\rangle = 0, \quad J_z|-j\rangle = -j|-j\rangle. \quad (16)$$

The eigenvalues range from j to $-j$ in steps of 1: $j, j-1, \dots, -j+1, -j$. The number of states is:

$$(j) - (-j) + 1 = j + j + 1 = 2j + 1, \quad (17)$$

implying $2j$ is an integer, so j is a non-negative integer or half-integer ($j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$).

1.5 Step 5: Use the Casimir Operator

Define the Casimir operator:

$$J^2 = J_x^2 + J_y^2 + J_z^2. \quad (18)$$

Express it using ladder operators:

$$J_x = \frac{J_+ + J_-}{2}, \quad J_y = \frac{J_+ - J_-}{2i}. \quad (19)$$

Compute:

$$J_x^2 = \left(\frac{J_+ + J_-}{2} \right)^2 = \frac{1}{4}(J_+^2 + J_+J_- + J_-J_+ + J_-^2), \quad (20)$$

$$J_y^2 = \left(\frac{J_+ - J_-}{2i} \right)^2 = \frac{1}{-4} (J_+^2 - J_+ J_- - J_- J_+ + J_-^2), \quad (21)$$

$$J_x^2 + J_y^2 = \frac{1}{4} (J_+^2 + J_+ J_- + J_- J_+ + J_-^2) - \frac{1}{4} (J_+^2 - J_+ J_- - J_- J_+ + J_-^2) = \frac{1}{2} (J_+ J_- + J_- J_+). \quad (22)$$

Thus:

$$J^2 = J_z^2 + \frac{1}{2} (J_+ J_- + J_- J_+). \quad (23)$$

Since $[J_+, J_-] = 2J_z$, we have $J_+ J_- = J_- J_+ + 2J_z$, so:

$$J^2 = J_z^2 + \frac{1}{2} (J_+ J_- + J_- J_+ - 2J_z + 2J_z) = J_z^2 + J_- J_+ + J_z. \quad (24)$$

For the highest weight vector:

$$J^2 |j\rangle = (J_z^2 + J_- J_+ + J_z) |j\rangle = (j^2 + J_- (0) + j) |j\rangle = (j^2 + j) |j\rangle = j(j+1) |j\rangle. \quad (25)$$

Since J^2 commutes with J_z, J_+, J_- , it is a scalar in an irreducible representation: $J^2 = j(j+1)I$. This confirms the eigenvalues $m = -j, -j+1, \dots, j$.

1.6 Step 6: Determine Coefficients

In a unitary representation, J_x, J_y, J_z are Hermitian, so $J_+^\dagger = J_-$. Assume orthonormal basis $\langle m | m' \rangle = \delta_{mm'}$:

$$J_+ |m\rangle = c_m |m+1\rangle, \quad J_- |m\rangle = d_m |m-1\rangle. \quad (26)$$

Since $\langle m | J_- | m+1 \rangle = \langle m+1 | J_+ | m \rangle^*$, compute:

$$||J_+ |m\rangle||^2 = \langle m | J_- J_+ | m \rangle = \langle m | (J_+ J_- - 2J_z) | m \rangle = c_{m-1} d_m - 2m, \quad (27)$$

$$||J_+ |m\rangle||^2 = |c_m|^2. \quad (28)$$

Thus, $|c_m|^2 = c_{m-1} d_m - 2m$. For $m = j$, $J_+ |j\rangle = 0$, so $c_j = 0$. For $m = -j$, $J_- | -j \rangle = 0$, so $d_{-j} = 0$. Solving recursively (standard in quantum mechanics) gives:

$$J_+ |m\rangle = \sqrt{j(j+1) - m(m+1)} |m+1\rangle, \quad J_- |m\rangle = \sqrt{j(j+1) - m(m-1)} |m-1\rangle. \quad (29)$$

1.7 Step 7: Relate to SO(3) Group

The Lie algebra $\mathfrak{so}(3)$ is isomorphic to $\mathfrak{su}(2)$, and its representations are labeled by $j = 0, \frac{1}{2}, 1, \dots$, with dimension $2j + 1$. Exponentiating to the group, $\text{SU}(2)$ (simply connected) admits all j , but $\text{SO}(3) = \text{SU}(2)/\mathbb{Z}_2$ requires single-valued representations. For integer j , the representation is single-valued on $\text{SO}(3)$; for half-integer j , it is double-valued, corresponding to $\text{SU}(2)$. Thus, $\text{SO}(3)$ representations have $j = 0, 1, 2, \dots$

1.8 Final Answer

The irreducible representations of the $\text{SO}(3)$ group, derived from the $\mathfrak{so}(3)$ Lie algebra via the ladder operator approach, are labeled by integers $j = 0, 1, 2, \dots$, with dimension $2j + 1$, where J_z has eigenvalues $m = -j, \dots, j$, and $J^2 = j(j + 1)I$.

2 Angular Momentum Operators and Their Transformation for $j = 1$

In quantum mechanics, for a system with angular momentum quantum number $j = 1$, the angular momentum operators are represented in the basis $|1, 1\rangle$, $|1, 0\rangle$, and $|1, -1\rangle$. We start with the ladder operators and J_z , compute J_x and J_y , and then apply a unitary transformation to obtain transformed operators.

2.1 1. Ladder Operators and J_z

The ladder operators J_+ (raising) and J_- (lowering), along with the z -component J_z , are defined as follows:

- **Raising operator** J_+ raises the magnetic quantum number m by 1:

$$J_+ = \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix} \quad (30)$$

- **Lowering operator** J_- lowers the magnetic quantum number m by 1:

$$J_- = \begin{pmatrix} 0 & 0 & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{pmatrix} \quad (31)$$

- **z -component** J_z is diagonal with eigenvalues corresponding to m :

$$J_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \quad (32)$$

2.2 2. Computation of J_x and J_y

Using the ladder operators, we compute J_x and J_y as follows:

- **J_x is the average of the raising and lowering operators:** Using the relation $J_x = \frac{1}{2}(J_+ + J_-)$,

$$J_x = \begin{pmatrix} 0 & \frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & 0 & \frac{\sqrt{2}}{2} \\ 0 & \frac{\sqrt{2}}{2} & 0 \end{pmatrix} \quad (33)$$

- **J_y involves the imaginary unit:** Using $J_y = \frac{1}{2i}(J_+ - J_-)$, where $\frac{1}{i} = -i$,

$$J_y = \begin{pmatrix} 0 & -\frac{\sqrt{2}i}{2} & 0 \\ \frac{\sqrt{2}i}{2} & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & \frac{\sqrt{2}i}{2} & 0 \end{pmatrix} \quad (34)$$

These matrices, along with J_z , satisfy the angular momentum commutation relations, such as $[J_x, J_y] = iJ_z$.

2.3 3. Unitary Transformation Matrix U

To transform the operators into a basis where certain properties (e.g., real rotation matrices) are emphasized, we use the unitary matrix U :

$$U = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} & 0 \end{pmatrix} \quad (35)$$

This matrix is unitary, satisfying $U^\dagger U = I$, where $U^\dagger$ is the conjugate transpose of U .

2.4 4. Transformed Angular Momentum Operators

The transformed operators are computed using the similarity transformation $J'_i = U^\dagger J_i U$ for $i = x, y, z$:

- **Transformed J'_x :**

$$J'_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \quad (36)$$

- **Transformed J'_y :**

$$J'_y = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix} \quad (37)$$

- **Transformed J'_z :**

$$J'_z = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (38)$$

2.5 5. Verification with iJ'_i

To explore the properties of the transformed operators, consider iJ'_i :

- $iJ'_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$
- $iJ'_y = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$
- $iJ'_z = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

Each iJ'_i is real and skew-symmetric (i.e., $(iJ'_i)^T = -iJ'_i$), which is consistent with generators of real rotation matrices in $\text{SO}(3)$.

2.6 Computing the rotation matrices

In this section we use:

- $J_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$
- $J_y = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$
- $J_z = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

We have:

$$J_x^2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad (39)$$

$$J_y^2 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad (40)$$

$$J_z^2 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (41)$$

and:

$$J_x^2 + J_y^2 + J_z^2 = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix}. \quad (42)$$

We have $J_x^3 = J_x J_x^2 = -J_x$ and $J_x^4 = J_x^2 J_x^2 = -J_x^2$ and $J_x^5 = -J_x^3 = J_x$.
Now we compute:

$$\mathbf{R}_x(\theta) = e^{\theta \mathbf{J}_x} = \mathbf{I} + \theta \mathbf{J}_x + \frac{(\theta \mathbf{J}_x)^2}{2!} + \frac{(\theta \mathbf{J}_x)^3}{3!} + \dots = \quad (43)$$

$$= \mathbf{I} + \theta \mathbf{J}_x + \frac{\theta^2 \mathbf{J}_x^2}{2!} - \frac{\theta^3 \mathbf{J}_x}{3!} - \frac{\theta^4 \mathbf{J}_x^2}{4!} + \frac{\theta^5 \mathbf{J}_x}{5!} + \dots = \quad (44)$$

$$= \mathbf{I} + \mathbf{J}_x \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \right) + \mathbf{J}_x^2 \left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \dots \right) = \quad (45)$$

$$= \mathbf{I} + \mathbf{J}_x^2 + \mathbf{J}_x \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots \right) - \mathbf{J}_x^2 \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots \right) = \quad (46)$$

$$= \mathbf{I} + \mathbf{J}_x^2 + \mathbf{J}_x \sin \theta - \mathbf{J}_x^2 \cos \theta = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{pmatrix}. \quad (47)$$

That is the rotation matrix around the x -axis.

3 The Homogeneous Lorentz Group from the Ground Up

3.1 The $so(1, 3)$ Lie Algebra

3.1.1 Step 1: Define the $so(1, 3)$ Lie Algebra

The Lie algebra $so(1, 3)$ is the Lie algebra of the Lorentz group, which preserves the Minkowski metric in four-dimensional spacetime with signature $(1, -1, -1, -1)$. It is a six-dimensional real Lie algebra generated by three rotation generators J_x, J_y, J_z (or J_i for $i = 1, 2, 3$) and three boost generators K_x, K_y, K_z (or K_i), corresponding to rotations in spatial planes and boosts mixing time and space coordinates, respectively.

In the physics convention, where generators are Hermitian in unitary representations, the commutation relations are:

$$[J_i, J_j] = i\epsilon_{ijk}J_k, \quad (48)$$

$$[J_i, K_j] = i\epsilon_{ijk}K_k, \quad (49)$$

$$[K_i, K_j] = -i\epsilon_{ijk}J_k, \quad (50)$$

where $i, j, k = 1, 2, 3$ (with $J_1 = J_x$, etc.), ϵ_{ijk} is the Levi-Civita symbol ($\epsilon_{123} = 1$), and summation over repeated indices is implied. Explicitly:

- $[J_x, J_y] = iJ_z$, cyclic permutations,
- $[J_x, K_y] = iK_z$, etc.,
- $[K_x, K_y] = -iJ_z$, etc.

These relations reflect that J_i generate an $so(3)$ subalgebra (rotations), while K_i introduce the non-compact nature of $so(1, 3)$ due to the negative sign in the boost commutator.

3.1.2 Step 2: Separate the Algebra Using a Complex Trick

To simplify the structure, define new generators:

$$A_i = \frac{J_i + iK_i}{2}, \quad B_i = \frac{J_i - iK_i}{2}, \quad (51)$$

for $i = 1, 2, 3$. These are complex linear combinations of the real generators J_i and K_i . Compute their commutation relations to reveal the algebra's structure.

Commutator $[A_i, A_j]$:

$$A_i = \frac{J_i + iK_i}{2}, \quad A_j = \frac{J_j + iK_j}{2}, \quad (52)$$

$$[A_i, A_j] = \left[\frac{J_i + iK_i}{2}, \frac{J_j + iK_j}{2} \right] = \frac{1}{4}[J_i + iK_i, J_j + iK_j]. \quad (53)$$

Expand:

$$[J_i + iK_i, J_j + iK_j] = [J_i, J_j] + i[J_i, K_j] + i[K_i, J_j] + i^2[K_i, K_j], \quad (54)$$

- $[J_i, J_j] = i\epsilon_{ijk}J_k$,
- $[J_i, K_j] = i\epsilon_{ijk}K_k$,
- $[K_i, J_j] = -[J_j, K_i] = -i\epsilon_{jik}K_k = i\epsilon_{ijk}K_k$ (since $\epsilon_{jik} = -\epsilon_{ijk}$),

- $[K_i, K_j] = -i\epsilon_{ijk}J_k$.

Substitute:

$$= i\epsilon_{ijk}J_k + i(i\epsilon_{ijk}K_k) + i(i\epsilon_{ijk}K_k) + (-1)(-i\epsilon_{ijk}J_k), \quad (55)$$

$$= i\epsilon_{ijk}J_k - \epsilon_{ijk}K_k - \epsilon_{ijk}K_k + i\epsilon_{ijk}J_k, \quad (56)$$

$$= 2i\epsilon_{ijk}J_k - 2\epsilon_{ijk}K_k, \quad (57)$$

$$= 4i\epsilon_{ijk}(J_k + iK_k)/2, \quad (58)$$

$$= 4i\epsilon_{ijk}A_k, \quad (59)$$

so:

$$[A_i, A_j] = \frac{1}{4}(4i\epsilon_{ijk}A_k) = i\epsilon_{ijk}A_k. \quad (60)$$

Commutator $[B_i, B_j]$:

$$B_i = \frac{J_i - iK_i}{2}, \quad B_j = \frac{J_j - iK_j}{2}, \quad (61)$$

$$[B_i, B_j] = \frac{1}{4}[J_i - iK_i, J_j - iK_j], \quad (62)$$

$$= \frac{1}{4}([J_i, J_j] - i[J_i, K_j] - i[K_i, J_j] + i^2[K_i, K_j]), \quad (63)$$

$$= \frac{1}{4}(i\epsilon_{ijk}J_k - i(i\epsilon_{ijk}K_k) - i(i\epsilon_{ijk}K_k) + (-1)(-i\epsilon_{ijk}J_k)), \quad (64)$$

$$= \frac{1}{4}(i\epsilon_{ijk}J_k + \epsilon_{ijk}K_k + \epsilon_{ijk}K_k + i\epsilon_{ijk}J_k), \quad (65)$$

$$= \frac{1}{4}(2i\epsilon_{ijk}J_k + 2\epsilon_{ijk}K_k) = i\epsilon_{ijk}\frac{J_k - iK_k}{2} = i\epsilon_{ijk}B_k. \quad (66)$$

Commutator $[A_i, B_j]$:

$$[A_i, B_j] = \left[\frac{J_i + iK_i}{2}, \frac{J_j - iK_j}{2} \right] = \frac{1}{4}[J_i + iK_i, J_j - iK_j], \quad (67)$$

$$= \frac{1}{4}([J_i, J_j] - i[J_i, K_j] + i[K_i, J_j] - i^2[K_i, K_j]), \quad (68)$$

$$= \frac{1}{4}(i\epsilon_{ijk}J_k - i(i\epsilon_{ijk}K_k) + i(i\epsilon_{ijk}K_k) - (-1)(-i\epsilon_{ijk}J_k)), \quad (69)$$

$$= \frac{1}{4}(i\epsilon_{ijk}J_k + \epsilon_{ijk}K_k - \epsilon_{ijk}K_k - i\epsilon_{ijk}J_k) = 0. \quad (70)$$

Thus:

$$[A_i, A_j] = i\epsilon_{ijk}A_k, \quad [B_i, B_j] = i\epsilon_{ijk}B_k, \quad [A_i, B_j] = 0. \quad (71)$$

Each set $\{A_i\}$ and $\{B_i\}$ satisfies the $su(2)$ commutation relations, and they commute with each other, suggesting $so(1, 3)$ is related to $su(2) \oplus su(2)$ via complexification.

3.1.3 Step 3: Interpret the Algebra Structure

Invert the definitions:

$$J_i = A_i + B_i, \quad K_i = -i(A_i - B_i). \quad (72)$$

The $so(1, 3)$ algebra, with real generators J_i, K_i , maps to two commuting $su(2)$ algebras via complex generators A_i, B_i . The complexification of $so(1, 3)$, denoted $so(1, 3)$, is isomorphic to $sl(2, \mathbb{C})$, and as a real Lie algebra, $so(1, 3)$ has dimension 6, matching $sl(2, \mathbb{C})$ over the reals. This trick reveals that representations can be constructed from $su(2)$ representations.

3.2 Representations of $so(1, 3)$

3.2.1 Step 4: Label Representations Using $su(2)$

Since A_i and B_i generate two independent $su(2)$ algebras, representations of $so(1, 3)$ are labeled by pairs (j_1, j_2) , where j_1 is the spin of the $su(2)$ algebra generated by A_i , and j_2 is the spin of the B_i algebra. From $su(2)$, $j_1, j_2 = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$, and the representation space is the tensor product of a $(2j_1 + 1)$ -dimensional space and a $(2j_2 + 1)$ -dimensional space, with dimension:

$$(2j_1 + 1)(2j_2 + 1). \quad (73)$$

In this representation:

- A_i acts as $J_i^{(j_1)} \otimes I$,

- B_i acts as $I \otimes J_i^{(j_2)}$,
- $J_i = A_i + B_i = J_i^{(j_1)} \otimes I + I \otimes J_i^{(j_2)}$,
- $K_i = -i(A_i - B_i) = -i(J_i^{(j_1)} \otimes I - I \otimes J_i^{(j_2)})$,

where $J_i^{(j)}$ are the $su(2)$ generators in the spin- j representation.

3.2.2 Step 5: Identify the Vector Representation

The homogeneous Lorentz group $SO(1, 3)$ acts on 4D Minkowski space via the four-vector representation, which is four-dimensional. Find (j_1, j_2) such that:

$$(2j_1 + 1)(2j_2 + 1) = 4. \quad (74)$$

- If $j_1 = 0, j_2 = \frac{3}{2}$: $(1)(4) = 4$ (but typically symmetric in physics),
- If $j_1 = \frac{1}{2}, j_2 = \frac{1}{2}$: $(2)(2) = 4$.

The $(1/2, 1/2)$ representation is the standard four-vector representation of $so(1, 3)$, corresponding to fields transforming as four-vectors (e.g., the electromagnetic potential).

3.3 The Homogeneous Lorentz Group $SO(1, 3)$

3.3.1 Step 6: Relate to the Group

The Lie algebra $so(1, 3)$ corresponds to the Lorentz group $SO(1, 3)$, which is non-compact, and its double cover is $SL(2, \mathbb{C})$. Finite-dimensional representations are not unitary, but the (j_1, j_2) labeling applies, with $SO(1, 3)$ transformations obtained by exponentiating the generators.

3.4 Transformation Matrices in the $(1/2, 1/2)$ Representation

3.4.1 Step 7: Construct Generators in $(1/2, 1/2)$

For $j_1 = 1/2, j_2 = 1/2$, the basis is $|m_1, m_2\rangle$ with $m_1, m_2 = \pm 1/2$, so:

- $|1/2, 1/2\rangle, |1/2, -1/2\rangle, |-1/2, 1/2\rangle, |-1/2, -1/2\rangle$.

Using $J_z^{(1/2)} = \sigma_z/2$, etc., where σ_i are Pauli matrices:

- $A_z = (\sigma_z/2) \otimes I = \text{diag}(1/2, 1/2, -1/2, -1/2)$,
- $B_z = I \otimes (\sigma_z/2) = \text{diag}(1/2, -1/2, 1/2, -1/2)$,
- $J_z = A_z + B_z = \text{diag}(1, 0, 0, -1)$,
- $K_z = -i(A_z - B_z) = -i\text{diag}(0, 1, -1, 0)$.

Compute other generators similarly (basis ordered as above).

3.4.2 Step 8: Adjust to Standard Basis

In the standard vector representation, J_i and K_i act on (t, x, y, z) :

- $J_z = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$

- $K_z = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$

3.4.3 Step 9: Exponentiate Generators

- **Rotation around z-axis:**

In standard basis, $e^{\theta J_z} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$

- **Boost along z-axis:**

$$e^{\zeta K_z} = \begin{pmatrix} \cosh \zeta & 0 & 0 & \sinh \zeta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh \zeta & 0 & 0 & \cosh \zeta \end{pmatrix}.$$

These are the standard Lorentz transformation matrices, derived via exponentiation in the adjusted basis.

3.5 Transformation of Lorentz Group Generators

In this section, we derive the standard real generators of the Lorentz group in the four-vector representation, starting from the $(1/2, 1/2)$ representation. This process involves a change of basis using a unitary matrix U followed by a rescaling by $-i$ to obtain real-valued generators.

3.5.1 Original Generators in the $(1/2, 1/2)$ Representation

The $(1/2, 1/2)$ representation arises from the tensor product of two spin-1/2 representations, typically constructed using Pauli matrices. The original generators consist of rotation generators J_i and boost generators K_i , defined as follows:

$$\begin{aligned}
J_x &= \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}, \\
J_y &= \begin{pmatrix} 0 & -\frac{i}{2} & -\frac{i}{2} & 0 \\ \frac{i}{2} & 0 & 0 & -\frac{i}{2} \\ \frac{i}{2} & 0 & 0 & -\frac{i}{2} \\ 0 & \frac{i}{2} & \frac{i}{2} & 0 \end{pmatrix}, \\
J_z &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \\
K_x &= \begin{pmatrix} 0 & \frac{i}{2} & -\frac{i}{2} & 0 \\ \frac{i}{2} & 0 & 0 & -\frac{i}{2} \\ -\frac{i}{2} & 0 & 0 & \frac{i}{2} \\ 0 & -\frac{i}{2} & \frac{i}{2} & 0 \end{pmatrix}, \\
K_y &= \begin{pmatrix} 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ -\frac{1}{2} & 0 & 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} & 0 \end{pmatrix}, \\
K_z &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.
\end{aligned}$$

These matrices are Hermitian, ensuring a unitary representation suitable for quantum mechanics, and they satisfy the Lorentz algebra with commutation relations involving factors of i .

3.5.2 Change of Basis Matrix U

To transform these generators into the standard four-vector representation, we define a change of basis using a unitary matrix U . This matrix is constructed from basis vectors corresponding to the four-vector components (t, x, y, z) :

$$\begin{aligned}
|t\rangle &= i \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}, \\
|x\rangle &= i \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ -\frac{1}{\sqrt{2}} \end{pmatrix}, \\
|y\rangle &= \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \\
|z\rangle &= -i \begin{pmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}.
\end{aligned}$$

The matrix U is formed by arranging these vectors as columns:

$$U = \begin{pmatrix} 0 & \frac{i\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ \frac{i\sqrt{2}}{2} & 0 & 0 & -\frac{i\sqrt{2}}{2} \\ -\frac{i\sqrt{2}}{2} & 0 & 0 & \frac{i\sqrt{2}}{2} \\ 0 & -\frac{i\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \end{pmatrix}. \quad (75)$$

This matrix is unitary, satisfying $U^\dagger U = I$, where $U^\dagger$ is the conjugate transpose of U , ensuring that the transformation preserves the algebraic structure.

3.5.3 Transformation and Rescaling by $-i$

We apply a similarity transformation to the original generators using U :

$$J'_i = U^\dagger J_i U, \quad K'_i = U^\dagger K_i U. \quad (76)$$

The resulting matrices J'_i and K'_i are purely imaginary due to the properties of the original generators and U . To match the standard real four-vector representation used in classical relativity, we rescale by multiplying by $-i$:

$$J_i^{\text{real}} = -iJ'_i, \quad K_i^{\text{real}} = -iK'_i. \quad (77)$$

This step converts the Hermitian, unitary representation into a real, anti-symmetric one, aligning with the conventions of the Lorentz group in four-vector form.

3.5.4 Final Real Generators

After applying the transformation and rescaling, the final real generators are:

$$J_x^{\text{real}} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

$$J_y^{\text{real}} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix},$$

$$J_z^{\text{real}} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$K_x^{\text{real}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$K_y^{\text{real}} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

$$K_z^{\text{real}} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

These generators are real and antisymmetric, matching the standard form of the Lorentz group generators in the four-vector representation, where J_i^{real} generate rotations and K_i^{real} generate boosts.

3.6 Final Summary

The $so(1,3)$ algebra, via the A and B trick, is represented as (j_1, j_2) , with the four-vector representation being $(1/2, 1/2)$. The transformation matrices for $SO(1,3)$ in this representation are the standard Lorentz transformations, obtained by exponentiating J_i and K_i after basis adjustment.